

Excel – Finance Case Study

FinServe Lending Co. — Loan Portfolio Analysis

FinServe Lending Co. is a mid-sized lending company operating across five regions in India — North, South, East, West, and Central — with 25 branches in cities including Delhi, Mumbai, Bangalore, Chennai, and Kolkata. Over the past three years, the company has disbursed loans to over 1,200 customers across a range of products: home loans, vehicle loans, education loans, personal loans, and business expansion loans.

Last quarter, FinServe's CFO, Ms. Priya Nair, noticed a sharp rise in late repayments and defaults across multiple branches. The Risk Team flagged that several high-risk customers may have slipped through without proper credit review. At the same time, management has no single view of the loan portfolio — data sits in scattered branch files, making it impossible to see the full picture.

Ms. Nair has asked the Analytics Team to step in. She wants a clean, reliable analysis of the loan portfolio — who are the risky borrowers, which branches are overexposed, and how repayment is tracking across the business. She has also asked for a simple dashboard she can open every month to monitor portfolio health at a glance.

You are the Excel Analyst on this BA team. The raw data has been extracted from the system and handed to you. It is not clean. Your job is to fix it, analyse it, and present it.

What Needs to Be Done

By the end of this session, you will answer the following business questions for FinServe's management:

- How much total loan amount has been disbursed, and what is the average loan size across regions?
- Which region carries the highest outstanding loan balance right now?
- How many borrowers fall into each risk category — Low, Medium, High, and Very High Risk?
- Which employment type has the most defaults and late payments?
- Which branches are carrying the most exposure, and which ones have the most high-risk customers?
- How has the volume of loan applications changed month by month?

To answer these questions, you will clean the data, add calculated columns, classify borrowers by risk, build pivot analysis, and put together a summary dashboard.

The Dataset

File: FinServe_Loan_Portfolio_Dataset.xlsx | Sheet: Loan_Data | Rows: 1,215

Column	What It Contains
Loan_ID	Unique reference number for each loan
Customer_Name	Full name of the borrower
Age	Age of the borrower
Employment_Type	Salaried / Self-Employed / Business Owner / Freelancer / Government Employee
Monthly_Income	Monthly income in Indian Rupees
Loan_Amount	Total loan sanctioned
Loan_Term_Months	Repayment period in months
Interest_Rate	Annual interest rate (%)
Loan_Purpose	Reason for the loan
Credit_Score	Borrower's credit score (300–900). Some values are missing.
Application_Date	Date the loan was applied for
Approval_Date	Date the loan was approved
Loan_Status	Approved / Rejected / Pending
EMI	Monthly repayment amount
Region	North / South / East / West / Central
Branch	Branch name (some entries are inconsistently formatted)
Repayment_Status	On Track / Late / Defaulted / Closed / Grace Period
Late_Payments	Number of late payments made so far
Outstanding_Balance	Current balance remaining on the loan

Reference Tables

The Excel file also contains a sheet called Reference_Tables with two lookup tables you will use during the analysis.

Risk Band Mapping

Credit Score Range	Risk Category	Risk Code	Action
750 – 900	Low Risk	LR	Approve – Standard Terms
650 – 749	Medium Risk	MR	Approve – With Conditions
550 – 649	High Risk	HR	Review – Senior Approval
300 – 549	Very High Risk	VHR	Reject or Collateral Required
Not Available	Unclassified	UC	Pending Review

Branch Code Reference

The second table lists all 25 branches with their Branch Code, Region, and Branch Manager name. Learners will use this table to practise XLOOKUP and INDEX+MATCH.

Tasks

Work through these tasks in order. Each section builds on the previous one.

Section 1 — Set Up and Clean the Data

Before any analysis can happen, the data needs to be in good shape. This section covers everything from converting the data into a proper Excel Table to fixing the inconsistencies that were left in on purpose.

1. Convert the dataset into an Excel Table using Ctrl + T. Name the table LoanData.
2. Find and remove the duplicate Loan_ID records. Note how many duplicates you removed.
3. The Customer_Name column has extra spaces at the start and end of some names. Use TRIM to fix this, then paste the cleaned values back into the column.
4. Employment_Type has inconsistent casing — some entries are in all caps, others are all lowercase. Use PROPER to standardise them, then paste the values back.
5. Loan_Purpose has hidden non-breaking space characters in some rows. Use a combination of TRIM, CLEAN, and SUBSTITUTE with CHAR(160) to remove them.
6. Credit_Score has blank cells for some borrowers. Use an IF formula to fill these with the text Not Available, then paste the values back.
7. Application_Date has two different date formats in the same column. Use Text to Columns to fix the non-standard entries so all dates are consistent.
8. Branch has entries like delhi connaught and MUMBAI-ANDHERI instead of Delhi-Connaught. Use SUBSTITUTE and PROPER to standardise these, then paste back.

Section 2 — Add Calculated Columns

Now that the data is clean, add these new columns in the Task_Workspace sheet. These will be used in the analysis and risk classification.

9. Days to Approval: Calculate how many days it took from Application_Date to Approval_Date.
10. Loan Age in Months: Using DATEDIF and TODAY, calculate how many months ago each loan was approved.
11. Maturity Date: Use EDATE to calculate when each loan will mature based on Approval_Date and Loan_Term_Months. Then add a second column called Is_Matured — if today is past that date, return Yes, otherwise No.
12. DTI Ratio: Calculate Debt-to-Income ratio as $(EMI \div Monthly_Income) \times 100$. Round to 2 decimal places. Then add a DTI Flag column — above 40 is High DTI Risk, between 25 and 40 is Moderate DTI, and below 25 is Low DTI. Use IFS for this.
13. Risk Category: Using an IFS formula and the Risk Band Mapping table, classify each borrower as Low Risk, Medium Risk, High Risk, Very High Risk, or Unclassified based on their Credit_Score.
14. Risk Code: Use XLOOKUP to look up the Risk_Category you just created and return the corresponding Risk Code from the reference table.
15. Branch Manager: Use INDEX + MATCH to look up each loan's Branch and return the name of the Branch Manager from the reference table.

Section 3 — Explore and Highlight the Data

Use sorting, filtering, and conditional formatting to start finding patterns in the data.

16. Sort the data by Outstanding_Balance from largest to smallest. Which customer has the highest balance?
17. Filter Repayment_Status to show only Defaulted loans. How many are there? Now add a second filter to show defaults from the North region only.
18. Apply conditional formatting on the Risk_Category column — Very High Risk in red, High Risk in orange, Medium Risk in yellow, Low Risk in green.
19. Apply a Data Bar to the Late_Payments column so the volume of late payments is visually obvious at a glance.

Section 4 — Pivot Table Analysis

Create a new sheet called Pivot_Analysis and build the following pivot tables to answer the business questions.

20. Loan Exposure by Region: Total Loan_Amount and Outstanding_Balance per Region. Which region has the highest total exposure?

21. Risk Distribution: Count of loans and total Loan_Amount by Risk_Category. What percentage of the portfolio is High Risk or Very High Risk?
22. Repayment by Employment Type: Cross-tab of Employment_Type vs Repayment_Status with Count of Loan_ID as values. Which employment type has the most defaults?
23. Branch Drill-Down: Nest Branch under Region in the rows. Collapse all regions and expand one at a time. Double-click a branch total to see the underlying detail — this is pivot drill-down.
24. Monthly Trend: Use Application_Date in rows and group it by Month and Year. Show Count of Loan_ID and Sum of Loan_Amount. Filter to Approved loans only. Which month had the highest approvals?

Section 5 — Charts

Create a new sheet called Charts. Build one chart from each pivot table above.

25. From the Region pivot — a Clustered Column chart showing total loan amount by region.
26. From the Branch pivot — a horizontal Bar chart showing the top 10 branches by loan exposure.
27. From the Monthly Trend pivot — a Line chart with markers showing how application volumes have moved month by month.
28. From the Risk Distribution pivot — a Pie or Doughnut chart showing the percentage split by risk category.
29. Chart Decision: You need to show Repayment Status by Region to management. Would you use a Clustered Column chart or a Stacked Bar chart? Build both, then write 2–3 sentences explaining which one you chose and why. There is no single right answer — the reasoning matters.

Section 6 — The Dashboard

The final deliverable is a one-screen management dashboard on the Dashboard_Shell sheet. Refer to the layout guide at the end of this document.

30. Add Pivot Chart 1 — Loan Distribution by Region (Column chart) in the top-left area.
31. Add Pivot Chart 2 — Portfolio Risk Segmentation (Doughnut chart) in the top-right area.
32. Add Pivot Chart 3 — Repayment Performance by Employment Type (Stacked Bar) in the lower area.
33. Add a Region slicer and a Loan_Status slicer. Connect both slicers to all three pivot charts using Report Connections so clicking a slicer updates every chart at once.
34. Add a Timeline filter for Application_Date at the bottom of the dashboard. Set it to show by Months.
35. Hide gridlines on the dashboard sheet. Add a title at the top: FinServe Lending Co. | Loan Portfolio Dashboard. Remove the field buttons from all pivot charts.

Dashboard Layout

The Dashboard Shell sheet already has placeholder labels showing where each element should go. Use this as your guide:

FinServe Lending Co. Loan Portfolio Dashboard		
Chart 1 Loan Distribution by Region (Clustered Column)	Chart 2 Risk Segmentation (Doughnut)	Slicer Region Slicer Loan Status
Chart 3 Repayment Performance by Employment Type (Stacked Bar)		(empty space)
Timeline — Application_Date (filter by Month / Quarter / Year)		

Keep all pivot tables on the Pivot_Analysis sheet. The dashboard should be clean — no raw data, no pivot ros, just the charts, slicers, and timeline visible.